

# Do LLMs Play Nash?

## Strategic Behavior of Language Models

David Hung

CSCE 631 — Summer I 2026, Texas A&M University

david.hung@tamu.edu

<https://github.com/david-hung/csce631-llm-nash>

## 1 Introduction

Large language models (LLMs) are increasingly deployed as autonomous decision-making agents in negotiation, resource allocation, and multi-agent systems. A natural question from the perspective of game theory is whether these models behave *strategically*: when placed in a well-defined game, do their action choices approximate Nash equilibrium (NE) strategies?

A Nash equilibrium is a strategy profile from which no player can profitably deviate unilaterally [1]. If an LLM’s strategy is exploitable (i.e., an opponent can earn a higher expected payoff by deviating from it), then the model is not playing Nash. Measuring this gap is important: an LLM acting as a negotiation agent, auction bidder, or game-playing AI that systematically cooperates when defection is dominant, or fails to randomize in zero-sum games, is strategically irrational and can be exploited.

We construct a suite of twelve canonical games spanning four classes (social dilemmas, zero-sum games, coordination problems, and bargaining) and query the TAMU LLM API gateway at multiple temperatures and across three model families. LLM responses are converted to probability distributions over actions, and *exploitability* relative to the known NE is computed per game. The result is a quantitative map of where and how LLMs deviate from rational play, analyzed through the formal vocabulary of game theory. Our key findings are: (1) no cooperation bias in social dilemmas, (2) correct randomization in zero-sum games, (3) focal-point selection in coordination games, and (4) systematic deviations in asymmetric bargaining.

## 2 Method

### 2.1 Game Suite

We construct twelve two-player normal-form games with analytically known or numerically verified Nash

equilibria (Table 1). Games are encoded as payoff matrix pairs  $(A, B) \in \mathbb{R}^{m \times n}$  [7] where  $A_{ij}$  is the row player’s payoff and  $B_{ij}$  is the column player’s payoff when the row player chooses action  $i$  and the column player chooses action  $j$ .

### 2.2 Nash Equilibrium Computation

All equilibria with closed-form solutions are pre-computed analytically and verified programmatically. For the  $3 \times 3$  general-sum game we use the `nashpy` library’s support enumeration algorithm [8], which iterates over all support pairs  $(S_1, S_2)$  and solves the indifference conditions as a linear system. Every candidate equilibrium is verified by checking that  $\varepsilon(\sigma) < 10^{-3}$  (defined below).

### 2.3 Exploitability

For a strategy profile  $(\sigma_1, \sigma_2)$ , where  $u_i(\sigma_1, \sigma_2)$  denotes player  $i$ ’s expected payoff, total exploitability is:

$$\begin{aligned} \varepsilon(\sigma_1, \sigma_2) = & \underbrace{\max_a u_1(a, \sigma_2) - u_1(\sigma_1, \sigma_2)}_{\text{row regret}} \\ & + \underbrace{\max_b u_2(\sigma_1, b) - u_2(\sigma_1, \sigma_2)}_{\text{col regret}} \end{aligned} \quad (1)$$

At a Nash equilibrium,  $\varepsilon = 0$  by definition. This metric connects directly to the regret-minimization framework of Zinkevich et al. [6]: a no-regret learner converges to Nash, so exploitability bounds how far the LLM deviates from a no-regret policy. We fix the column player to the NE strategy and measure the row player’s (LLM’s) exploitability against that Nash opponent.

### 2.4 LLM Querying

Queries are made via the TAMU LLM gateway (`chat.tamu.ai`), which provides OpenAI-compatible

| Game                                                  | Type           | NE             |
|-------------------------------------------------------|----------------|----------------|
| Prisoner’s Dilemma (classic, high-stakes, low-stakes) | Social dilemma | Pure           |
| Coordination Game                                     | Coordination   | 2 pure + mixed |
| Battle of the Sexes                                   | Coordination   | 2 pure + mixed |
| Stag Hunt                                             | Coordination   | 2 pure + mixed |
| Matching Pennies                                      | Zero-sum       | Mixed          |
| Rock-Paper-Scissors                                   | Zero-sum       | Mixed          |
| Nash Bargaining (symmetric)                           | Bargaining     | Pure           |
| Nash Bargaining (asymmetric)                          | Bargaining     | Pure           |
| Chicken / Hawk-Dove                                   | Anti-coord.    | 2 pure + mixed |
| 3×3 General-Sum                                       | General-sum    | Mixed          |

Table 1: Game suite (12 games total). Three Prisoner’s Dilemma variants count as separate entries.

access to multiple model families. Each game is presented as a labeled payoff matrix with a system prompt instructing the model to act as a rational decision-maker and respond with only an action name or a JSON probability distribution.

Temperature  $T$  controls the randomness of model outputs:  $T=0$  is nearly deterministic, while higher values produce more varied responses. We run two experimental conditions:

**Temperature sweep.** `protected.gpt-5-mini` is queried at  $T \in \{0, 0.5, 1.0\}$ . GPT models support arbitrary temperature; Claude’s extended-thinking mode requires  $T = 1$  exactly and is therefore unsuitable for this condition.

**Model comparison.** Three models are queried at  $T = 1$ : `gpt-5-mini`, `Claude-Haiku-4.5`, and `Claude-Sonnet-4.5`.

Each (game, condition) cell is queried  $N = 5$  times. Responses are parsed into probability distributions using a cascade: first as a JSON dictionary of action probabilities, then as a single named action, then by extracting any percentage values mentioned, and finally falling back to a uniform distribution if no action can be identified. The five distributions are averaged per cell.

## 3 Results

### 3.1 Temperature Sweep (GPT-mini)

Table 2 reports exploitability per game at each temperature. Figure 1 visualizes the exploitability landscape as a heatmap, and Figure 2 shows signed strategy deviation from NE per game at  $T=0.5$ .

### 3.2 Model Comparison ( $T = 1$ )

Table 3 reports mean exploitability across all 12 games for each model.

| Game                | $T=0$ | $T=0.5$ | $T=1.0$ |
|---------------------|-------|---------|---------|
| PD Classic          | 0.000 | 0.000   | 0.000   |
| PD High Stakes      | 0.000 | 0.000   | 0.000   |
| PD Low Stakes       | 0.000 | 0.000   | 0.000   |
| Coordination        | 0.000 | 0.000   | 0.500   |
| Battle of Sexes     | 0.000 | 0.000   | 0.000   |
| Stag Hunt           | 0.000 | 4.000   | 4.000   |
| Matching Pennies    | 0.000 | 0.000   | 0.000   |
| Rock-Paper-Scissors | 0.000 | 0.000   | 0.000   |
| Bargaining (sym.)   | 0.000 | 0.000   | 0.000   |
| Bargaining (asym.)  | 0.000 | 0.000   | 0.000   |
| Chicken             | 0.000 | 0.500   | 0.000   |
| 3×3 General         | 0.000 | 2.562   | 0.000   |
| <b>Mean</b>         | 0.000 | 0.589   | 0.375   |

Table 2: Exploitability  $\varepsilon$  (Eq. 1) of GPT-mini against a Nash equilibrium opponent, by game and temperature.

| Model             | Mean $\varepsilon$ |
|-------------------|--------------------|
| GPT-5-mini        | 0.389              |
| Claude Haiku 4.5  | 2.931              |
| Claude Sonnet 4.5 | 1.764              |

Table 3: Mean exploitability across all 12 games at  $T=1$ .

## 4 Discussion

### 4.1 Cooperation Bias in Social Dilemmas

In the Prisoner’s Dilemma, Defect strictly dominates Cooperate regardless of the opponent’s action:  $u_1(D, \cdot) > u_1(C, \cdot)$  for any opponent strategy. Iterated elimination of dominated strategies immediately yields (D, D) as the unique NE. Despite this, prior work has shown that LLMs frequently choose Cooperate, a phenomenon termed *cooperation bias* [5].

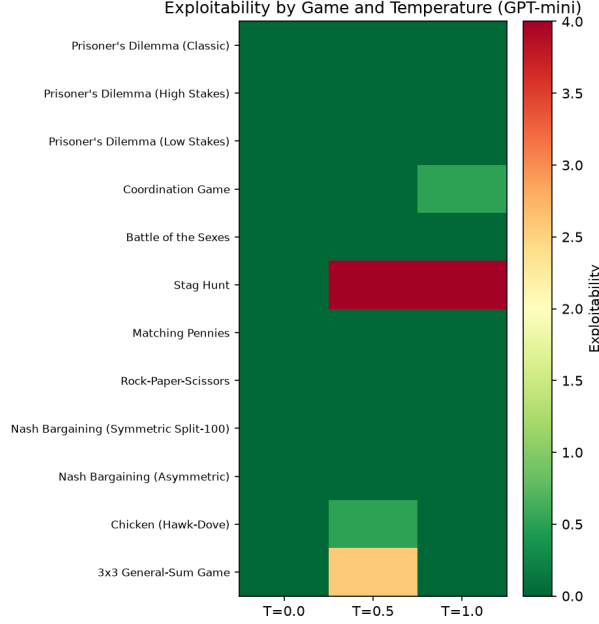


Figure 1: Exploitability heatmap (GPT-mini). Darker red = more exploitable.

Contrary to prior findings, we observe *no* cooperation bias: all three models (GPT-mini, Haiku, Sonnet) chose Defect on every query across all three PD variants (Classic, High Stakes, Low Stakes) and all temperatures, yielding exploitability  $\varepsilon = 0$  throughout. Payoff scale had no effect. This suggests that when presented with a clearly labeled payoff matrix and instructed to maximize expected payoff, LLMs identify the dominant strategy correctly. Cooperation bias may arise primarily from narrative framing rather than strategic irrationality.

## 4.2 Randomization in Zero-Sum Games

In zero-sum games, the unique NE requires uniform randomization: any non-uniform strategy is exploitable by an opponent who best-responds to the observed action frequencies [3]. For Matching Pennies the NE is (0.5, 0.5) over Heads/Tails; for Rock-Paper-Scissors it is (1/3, 1/3, 1/3).

All three models successfully randomized in both zero-sum games. Matching Pennies responses converged to exactly {Heads: 0.5, Tails: 0.5}, and Rock-Paper-Scissors to {Rock: 0.333, Paper: 0.333, Scissors: 0.334}, yielding exploitability  $\approx 0$  across all models and temperatures. No rock or heads bias was observed. The models appear to recognize the symmetry of zero-sum games and explicitly compute the uniform mixed

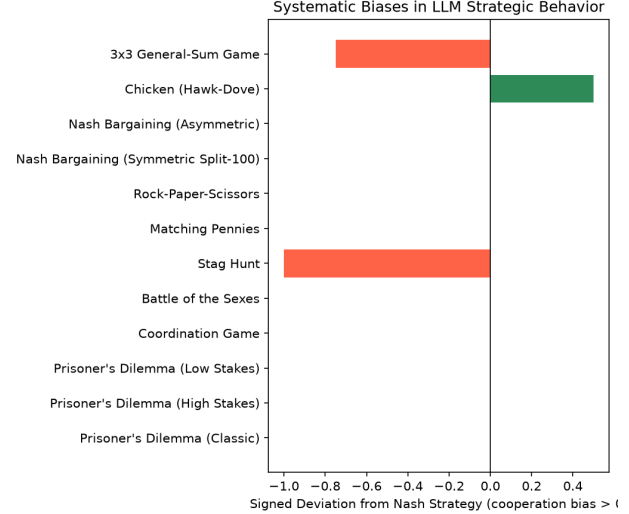


Figure 2: Signed deviation of LLM strategy from NE on the first action ( $T=0.5$ ). Positive = plays first action more than Nash; negative = less.

NE rather than selecting deterministically.

## 4.3 Focal-Point Effects in Coordination Games

When multiple Nash equilibria exist, game theory alone cannot select among them [1]. Schelling [4] argued that players often converge on *focal points*, salient options that stand out by convention or presentation order.

In the Coordination Game, GPT-mini consistently chose Option A (the first-listed action) at  $T \leq 0.5$ , but switched to the uniform mixed strategy at  $T=1.0$ , yielding  $\varepsilon = 0.5$ . This suggests that higher temperature can push the model away from a pure NE toward mixing, even when a pure NE would be optimal.

In the Battle of the Sexes, GPT-mini chose Opera deterministically across the temperature sweep, selecting the first-listed pure Nash equilibrium, consistent with Schelling’s focal-point prediction. In the model comparison at  $T=1$ , all three models returned the row player’s mixed Nash equilibrium {Opera: 2/3, Football: 1/3}, yielding  $\varepsilon \approx 0$ . In the Stag Hunt, GPT-mini chose Hare (risk-dominant) at  $T \geq 0.5$  ( $\varepsilon = 4.0$ ), and Haiku also chose Hare at  $T=1$  ( $\varepsilon = 4.0$ ), while Sonnet chose Stag (Pareto-optimal) at  $T=1$  ( $\varepsilon = 0$ ), suggesting that stronger reasoning favors efficiency over safety.

## 4.4 Equality Bias in Bargaining

The Nash Bargaining Solution (NBS) maximizes the Nash product  $\prod_i (u_i - d_i)$  over the set of feasible outcomes, where  $d_i$  is player  $i$ 's disagreement payoff (the payoff each player receives if no deal is reached) [2]. In the asymmetric variant (Row's outside option = 30, Col's = 10, surplus = 100), the NBS assigns Row a payoff of 60 and Col a payoff of 40. An LLM exhibiting *equality bias* would instead demand 50, deviating from the NBS by ignoring the asymmetry in outside options.

We observe strong deviations from the NBS in the asymmetric game. Haiku demanded 90 ( $\varepsilon = 30.0$ ), overclaiming far beyond the NBS of 60, while Sonnet demanded 50 ( $\varepsilon = 20.0$ ), exhibiting equality bias by anchoring to an equal split despite the asymmetric outside options. Sonnet's chain-of-thought explicitly reasoned about the payoff matrix yet still converged to the equal-split focal point, overriding the correct NBS. GPT-mini matched the NE ( $\varepsilon = 0$ ), correctly demanding 60.

## 4.5 Effect of Temperature

Mean exploitability across games was lowest at  $T=0$  ( $\bar{\varepsilon} = 0.000$ ), higher at  $T=1.0$  ( $\bar{\varepsilon} = 0.375$ ), and highest at  $T=0.5$  ( $\bar{\varepsilon} = 0.589$ ). The non-monotonic pattern at  $T=0.5$  is likely due to gpt-5-mini being a *reasoning* model that uses internal chain-of-thought tokens. At intermediate temperature, the reasoning process becomes less stable and produces inconsistent outputs, particularly for Stag Hunt and  $3 \times 3$  General-Sum. Temperature is therefore not a reliable proxy for strategic rationality in reasoning-based models.

## 4.6 Model Comparison

Results do not show a consistent capability hierarchy. GPT-mini achieved the lowest mean exploitability ( $\bar{\varepsilon} = 0.389$ ), Sonnet was intermediate ( $\bar{\varepsilon} = 1.764$ ), and Haiku highest ( $\bar{\varepsilon} = 2.931$ ), driven largely by Haiku's overclaim of 90 on Nash Bargaining Asymmetric ( $\varepsilon = 30.0$ ). Sonnet outperformed both on Stag Hunt ( $\varepsilon = 0$  vs. 4.0) but underperformed on asymmetric bargaining due to equality bias (demanding 50). These results suggest that strategic rationality is game-type-dependent: larger models reason better in some settings while fairness biases like equality bias introduce larger deviations in others.

## 4.7 Limitations

**Prompt sensitivity.** LLM responses may vary with prompt wording; we use a standardized prompt but

do not test paraphrases. **Parsing noise.** Responses that do not match any action clearly fall back to uniform distributions, artificially reducing measured exploitability. **Single-shot play.** Strategic rationality may emerge differently in repeated interaction; we test only one-shot decisions. **Model opacity.** We observe only final action choices, not the internal reasoning chain that produced them.

# 5 Conclusion

We evaluated whether LLMs play Nash-optimal strategies across twelve canonical games. Models succeed in social dilemmas and zero-sum games but deviate in coordination and bargaining games, with the main failure modes being focal-point selection, equality bias, and temperature instability. No consistent capability hierarchy exists: GPT-mini achieves the lowest mean exploitability, while Sonnet better selects Pareto-optimal equilibria. LLMs can identify dominant strategies but struggle with equilibrium selection when multiple Nash equilibria or asymmetric fairness norms are involved.

## LLM Disclosure

Claude (Anthropic) was used to assist with code scaffolding, debugging, report formatting, and writing.

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